

BUSA8090 - Data and Visualisation for Business Assignment 1

David Jones Marketing Campaign Strategy: Analytics and Database Design Report

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Part A – Database Design and Development

1.1 Entity Identification

Entity Name	Entity Description
CUSTOMER	Represents users who interact with advertisement and make purchases on the David Jones websites.
SOCIALMEDIAPLATFORM	Represents social media platforms such as Instagram, Facebook, X, where campaigns are conducted.
CAMPAIGN	Represents marketing campaigns such as EOFY sale that are used to promote products over a specific time period.
ADVERTISEMENT	Represents individual ads created under each campaign to engage customers.
IMPRESSION	Represents when a customer views an advertisement.
CLICK	Represents when a customer clicks on an advertisement which shows engagement.
ORDER	Represents purchases made by customer on the website.
ORDER ITEM	Represents individual products included in each order.
PRODUCT	Represents items sold by David Jones

1.2 Identify Attributes, Primary Keys and Foreign Keys

Entity Name	Attributes
CUSTOMER	customer_id(PK), customer_name, customer_email, customer_age, customer_gender
SOCIAL MEDIA PLATFORM	platform_id(PK), platform_name
CAMPAIGN	campaign_id(PK), platform_id(FK), campaign_name, start_date, end_date
ADVERTISEMENT	ad_id(PK), campaign_id(FK), ad_name, ad_type, posted_date
IMPRESSION	impression_id(PK), ad_id(FK), customer_id(FK), impression_time
CLICK	click_id(PK), ad_id(FK), customer_id(FK), source_url, click_time
ORDER	order_id(PK), customer_id(FK), order_date, order_type, order_status, payment_method, total_amount, shipping_cost
ORDER ITEM	Order_item_id(PK), order_id(FK), product_id(FK), quantity, unit_price
PRODUCT	Product_id(PK), product_name, brand_name, category, price

1.3 Relationship Definition and Cardinality Analysis

A social media platform hosts many campaigns, but each campaign is hosted by one social media platform (one to many).

A campaign creates many advertisements, but each advertisement is included in one campaign (one to many).

An advertisement generates many impressions, but each impression is generated by one advertisement (one to many).

A customer views many impressions, but each impression is viewed by one customer (one to many).

An advertisement receives many clicks, but each click is received by one advertisement (one to many).

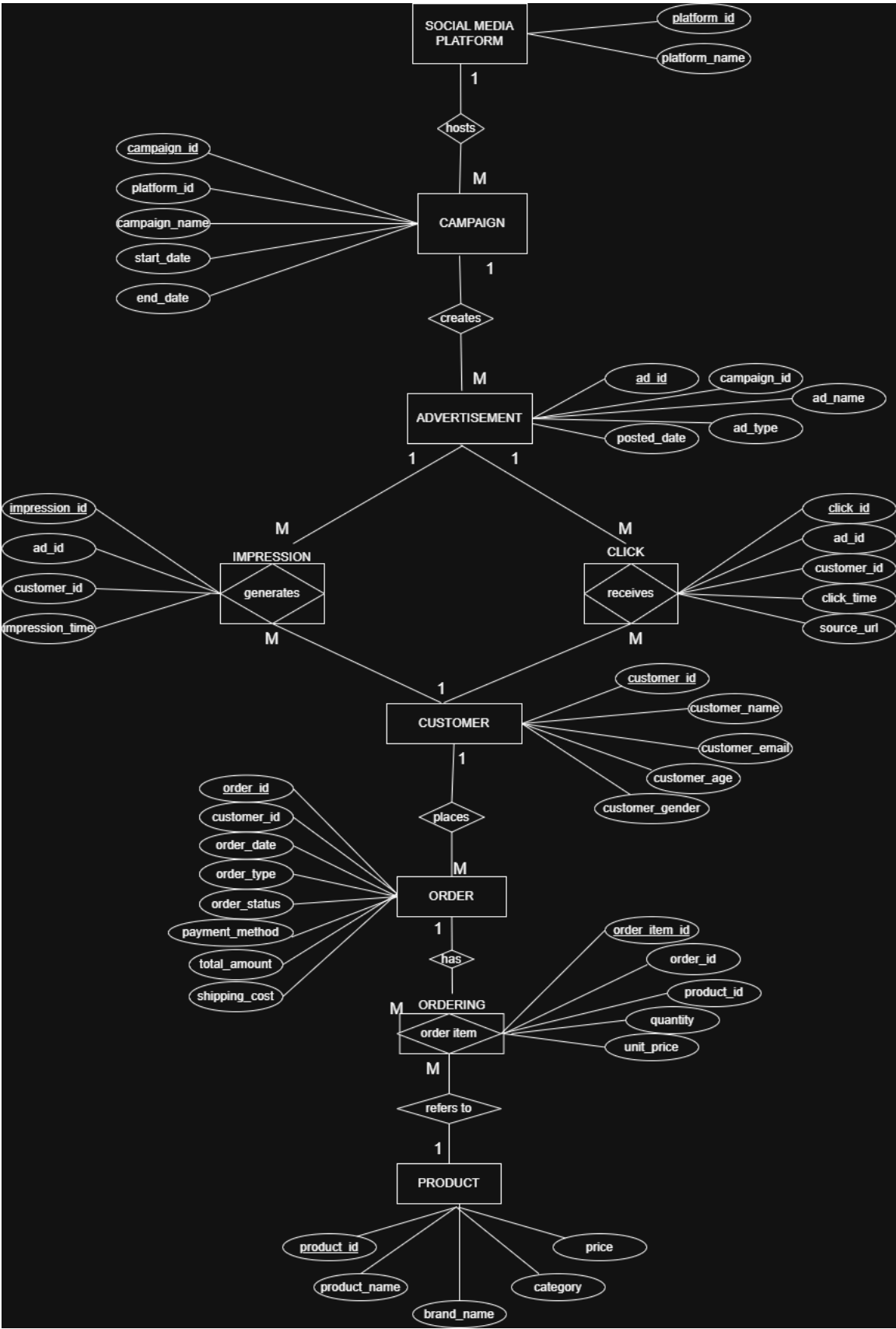
A customer performs many clicks, but each click is performed by one customer (one to many).

A customer can place many orders, but each order is placed by one customer (one to many)

An order contains many order items, but each order item is contained in one order (one to many)

A product appears in many order items, but each order item refers to one product (one to many)

1.4 ERD diagram



Note: If the ER diagram is not clearly visible in this report, a high-resolution version has also been included in the submitted ZIP file.

Database Design and Implementation:

The database was designed by first identifying the main entities involved in the online shopping and marketing process. These included Social Media Platform, Campaign, Advertisement, Impression, Click, Customer, Order, Order Item and Product. These entities were chosen to represent how a customer interacts with advertisements and eventually makes a purchase.

After identifying the entities, I defined the attributes for each table, such as unique IDs, names, dates and other relevant details. Primary keys were assigned to each table to uniquely identify records, and foreign keys were added to connect related tables. For example, `platform_id` connects SocialMediaPlatform to Campaign, `campaign_id` connects Campaign to Advertisement, and `ad_id` connects Advertisement to Impression and Click.

Next, I created the relationships between the entities based on the ER diagram. These relationships follow a one-to-many structure, such as one platform having many campaigns, and one customer placing many orders. To handle the many-to-many relationship between Order and Product, I created an associative entity such as Order Item table, which acts as a linking table.

The database was then implemented in MySQL Workbench. I created each table using SQL CREATE TABLE statements and defined the primary and foreign key constraints. After creating the tables, I inserted dummy data using INSERT INTO statements. The data was designed to include different platforms, campaigns, customers and transactions to support meaningful analysis.

Finally, I tested the database by running SQL queries to ensure that the relationships were working correctly and that the data could be used to generate insights on customer behaviour, marketing performance and sales.

1.5 Table implementation

```
USE david_jones;
```

```
CREATE TABLE Customer (  
    customer_id INT PRIMARY KEY,  
    customer_name VARCHAR(100),  
    customer_email VARCHAR(100),  
    customer_age INT,  
    customer_gender VARCHAR(20)  
);
```

```
CREATE TABLE SocialMediaPlatform (  
    platform_id INT PRIMARY KEY,  
    platform_name VARCHAR(50)  
);
```

```
CREATE TABLE Campaign (  
    campaign_id INT PRIMARY KEY,  
    platform_id INT,  
    campaign_name VARCHAR(100),  
    start_date DATE,  
    end_date DATE,  
    FOREIGN KEY (platform_id) REFERENCES SocialMediaPlatform(platform_id)  
);
```

```
CREATE TABLE Advertisement (  
    ad_id INT PRIMARY KEY,  
    campaign_id INT,  
    ad_name VARCHAR(100),  
    ad_type VARCHAR(50),  
    posted_date DATE,  
    FOREIGN KEY (campaign_id) REFERENCES Campaign(campaign_id)  
);
```

```
CREATE TABLE Impression (  
    impression_id INT PRIMARY KEY,  
    ad_id INT,  
    customer_id INT,  
    impression_time DATETIME,  
    FOREIGN KEY (ad_id) REFERENCES Advertisement(ad_id),  
    FOREIGN KEY (customer_id) REFERENCES Customer(customer_id)  
);
```

```
CREATE TABLE Click (  
    click_id INT PRIMARY KEY,  
    ad_id INT,  
    customer_id INT,  
    source_url VARCHAR(255),  
    click_time DATETIME,  
    FOREIGN KEY (ad_id) REFERENCES Advertisement(ad_id),  
    FOREIGN KEY (customer_id) REFERENCES Customer(customer_id)  
);
```

```
CREATE TABLE `Order` (  
    order_id INT PRIMARY KEY,  
    customer_id INT,  
    order_date DATE,  
    order_type VARCHAR(50),  
    order_status VARCHAR(50),  
    payment_method VARCHAR(50),  
    total_amount DECIMAL(10,2),  
    shipping_cost DECIMAL(10,2),  
    FOREIGN KEY (customer_id) REFERENCES Customer(customer_id)  
);
```

```
CREATE TABLE Product (  
    product_id INT PRIMARY KEY,  
    product_name VARCHAR(100),  
    brand_name VARCHAR(100),  
    category VARCHAR(50),  
    price DECIMAL(10,2)  
);
```

```
CREATE TABLE OrderItem (  
    order_item_id INT PRIMARY KEY,  
    order_id INT,  
    product_id INT,  
    quantity INT,  
    unit_price DECIMAL(10,2),  
    FOREIGN KEY (order_id) REFERENCES `Order` (order_id),  
    FOREIGN KEY (product_id) REFERENCES Product(product_id)
```

```
);
```

1.6 Dummy Data insertion

```
USE david_jones;
```

```
#Social Media Platform
```

```
INSERT INTO SocialMediaPlatform VALUES
```

```
(1, 'Instagram'),
```

```
(2, 'Facebook'),
```

```
(3, 'X');
```

```
#Customer
```

```
INSERT INTO Customer VALUES
```

```
(1,'John Smith','john@email.com',25,'Male'),
```

```
(2,'Sarah Lee','sarah@email.com',30,'Female'),
```

```
(3,'David Kim','david@email.com',28,'Male'),
```

```
(4,'Emily Brown','emily@email.com',35,'Female'),
```

```
(5,'Michael Tan','michael@email.com',40,'Male'),
```

```
(6,'Jessica Lim','jessica@email.com',22,'Female'),
```

```
(7,'Daniel Wong','daniel@email.com',27,'Male'),
```

```
(8,'Olivia Chen','olivia@email.com',31,'Female'),
```

```
(9,'Liam Patel','liam@email.com',29,'Male'),
```

```
(10,'Sophia White','sophia@email.com',33,'Female'),
```

```
(11,'Ethan Clark','ethan@email.com',26,'Male'),
```

```
(12,'Chloe Green','chloe@email.com',34,'Female'),
```

```
(13,'Noah Singh','noah@email.com',28,'Male'),
```

```
(14,'Ava Wilson','ava@email.com',32,'Female'),
```

(15,'Lucas Hall','lucas@email.com',24,'Male');

#Campaign

INSERT INTO Campaign VALUES

(1,1,'EOFY Sale','2026-05-01','2026-06-30'),
(2,1,'Summer Promo','2026-06-01','2026-07-01'),
(3,2,'Winter Sale','2026-06-01','2026-07-15'),
(4,2,'Clearance','2026-05-10','2026-06-20'),
(5,3,'Flash Sale','2026-05-15','2026-06-15'),
(6,3,'Weekend Deal','2026-06-05','2026-06-30'),
(7,1,'New Arrivals','2026-06-10','2026-07-10'),
(8,2,'Luxury Promo','2026-05-20','2026-06-25'),
(9,3,'Budget Deals','2026-06-01','2026-07-01'),
(10,1,'Holiday Promo','2026-06-15','2026-07-15');

#Advertisement

INSERT INTO Advertisement VALUES

(1,1,'EOFY Ad 1','Image','2026-05-01'),
(2,1,'EOFY Ad 2','Video','2026-05-05'),
(3,2,'Summer Ad 1','Image','2026-06-01'),
(4,2,'Summer Ad 2','Video','2026-06-02'),
(5,3,'Winter Ad 1','Image','2026-06-01'),
(6,3,'Winter Ad 2','Video','2026-06-03'),
(7,4,'Clearance Ad 1','Image','2026-05-10'),
(8,5,'Flash Ad 1','Video','2026-05-15'),
(9,6,'Weekend Ad','Image','2026-06-05'),
(10,7,'New Arrival Ad','Video','2026-06-10'),
(11,8,'Luxury Ad','Image','2026-05-20'),

```
(12,9,'Budget Ad','Image','2026-06-01'),  
(13,10,'Holiday Ad','Video','2026-06-15'),  
(14,4,'Clearance Ad 2','Video','2026-05-12'),  
(15,5,'Flash Ad 2','Image','2026-05-18');
```

#Product

INSERT INTO Product VALUES

```
(1,'Shirt','Nike','Clothing',50),  
(2,'Shoes','Adidas','Footwear',120),  
(3,'Bag','Gucci','Accessories',300),  
(4,'Watch','Rolex','Accessories',500),  
(5,'Jacket','Zara','Clothing',150),  
(6,'Hat','Puma','Accessories',40),  
(7,'Sneakers','Nike','Footwear',130),  
(8,'Dress','H&M','Clothing',80),  
(9,'Jeans','Levis','Clothing',90),  
(10,'Backpack','NorthFace','Accessories',110),  
(11,'Sunglasses','Rayban','Accessories',200),  
(12,'Coat','Uniqlo','Clothing',180),  
(13,'Boots','Timberland','Footwear',220),  
(14,'T-shirt','Adidas','Clothing',45),  
(15,'Scarf','Zara','Accessories',60);
```

#Impression

INSERT INTO Impression VALUES

```
(1,1,1,'2026-05-01 10:00:00'),  
(2,1,2,'2026-05-01 11:00:00'),  
(3,2,3,'2026-05-02 12:00:00'),
```

(4,2,4,'2026-05-02 13:00:00'),
(5,3,5,'2026-06-01 14:00:00'),
(6,3,6,'2026-06-01 15:00:00'),
(7,4,7,'2026-06-02 16:00:00'),
(8,4,8,'2026-06-02 17:00:00'),
(9,5,9,'2026-06-01 18:00:00'),
(10,5,10,'2026-06-01 19:00:00'),
(11,6,11,'2026-06-03 10:00:00'),
(12,6,12,'2026-06-03 11:00:00'),
(13,7,13,'2026-05-10 12:00:00'),
(14,7,14,'2026-05-10 13:00:00'),
(15,8,15,'2026-05-15 14:00:00'),
(16,8,1,'2026-05-15 15:00:00'),
(17,9,2,'2026-06-05 16:00:00'),
(18,9,3,'2026-06-05 17:00:00'),
(19,10,4,'2026-06-10 18:00:00'),
(20,10,5,'2026-06-10 19:00:00'),
(21,11,6,'2026-05-20 10:00:00'),
(22,11,7,'2026-05-20 11:00:00'),
(23,12,8,'2026-06-01 12:00:00'),
(24,12,9,'2026-06-01 13:00:00'),
(25,13,10,'2026-06-15 14:00:00'),
(26,13,11,'2026-06-15 15:00:00'),
(27,14,12,'2026-05-12 16:00:00'),
(28,14,13,'2026-05-12 17:00:00'),
(29,15,14,'2026-05-18 18:00:00'),
(30,15,15,'2026-05-18 19:00:00');

#Click

INSERT INTO Click VALUES

(1,1,1,'google.com','2026-05-01 10:05:00'),
(2,1,2,'google.com','2026-05-01 11:05:00'),
(3,2,3,'instagram.com','2026-05-02 12:05:00'),
(4,2,4,'instagram.com','2026-05-02 13:05:00'),
(5,3,5,'facebook.com','2026-06-01 14:05:00'),
(6,3,6,'facebook.com','2026-06-01 15:05:00'),
(7,4,7,'facebook.com','2026-06-02 16:05:00'),
(8,4,8,'facebook.com','2026-06-02 17:05:00'),
(9,5,9,'x.com','2026-06-01 18:05:00'),
(10,5,10,'x.com','2026-06-01 19:05:00'),
(11,6,11,'facebook.com','2026-06-03 10:05:00'),
(12,6,12,'facebook.com','2026-06-03 11:05:00'),
(13,7,13,'google.com','2026-05-10 12:05:00'),
(14,7,14,'google.com','2026-05-10 13:05:00'),
(15,8,15,'x.com','2026-05-15 14:05:00'),
(16,8,1,'x.com','2026-05-15 15:05:00'),
(17,9,2,'instagram.com','2026-06-05 16:05:00'),
(18,9,3,'instagram.com','2026-06-05 17:05:00'),
(19,10,4,'facebook.com','2026-06-10 18:05:00'),
(20,10,5,'facebook.com','2026-06-10 19:05:00'),
(21,11,6,'google.com','2026-05-20 10:05:00'),
(22,12,7,'instagram.com','2026-06-01 12:05:00'),
(23,13,8,'x.com','2026-06-15 14:05:00'),
(24,14,9,'google.com','2026-05-12 16:05:00'),

```
(25,15,10,'x.com','2026-05-18 18:05:00');
```

```
#Order
```

```
INSERT INTO `Order` VALUES
```

```
(1,1,'2026-05-02','Online','Completed','Card',170,10),  
(2,2,'2026-05-03','Online','Completed','PayPal',120,5),  
(3,3,'2026-05-04','Online','Pending','Card',300,15),  
(4,4,'2026-06-02','Online','Completed','Card',500,20),  
(5,5,'2026-06-03','Online','Completed','PayPal',150,10),  
(6,6,'2026-05-06','Online','Cancelled','Card',50,5),  
(7,7,'2026-06-04','Online','Completed','Card',120,10),  
(8,8,'2026-06-05','Online','Completed','PayPal',300,15),  
(9,9,'2026-05-21','Online','Completed','Card',500,20),  
(10,10,'2026-05-22','Online','Completed','Card',150,10),  
(11,11,'2026-06-01','Online','Completed','Card',200,10),  
(12,12,'2026-06-02','Online','Completed','PayPal',180,10),  
(13,13,'2026-06-03','Online','Pending','Card',90,5),  
(14,14,'2026-06-04','Online','Completed','Card',130,10),  
(15,15,'2026-06-05','Online','Completed','Card',220,15);
```

```
#OrderItem
```

```
INSERT INTO OrderItem VALUES
```

```
(1,1,1,2,50),(2,1,2,1,120),(3,2,2,1,120),  
(4,3,3,1,300),(5,4,4,1,500),(6,5,5,1,150),  
(7,6,1,1,50),(8,7,2,1,120),(9,8,3,1,300),  
(10,9,4,1,500),(11,10,5,1,150),(12,2,1,1,50),  
(13,3,2,1,120),(14,4,5,1,150),(15,5,3,1,300),
```

(16,11,6,1,40),(17,12,7,1,130),(18,13,8,1,80),
(19,14,9,1,90),(20,15,10,1,110),(21,11,11,1,200),
(22,12,12,1,180),(23,13,13,1,220),(24,14,14,1,45),
(25,15,15,1,60)

Part B – Business Analytics Report

1.7 Business concern 1

Name and Description:

Social Media Click Through Rate (CTR) Analysis

This concern analyses the effectiveness of social media platforms by measuring the click-through rate (CTR). CTR shows how many users who view an advertisement actually click on it. This helps the CMO understand which platform is more effective in converting views into engagement and supports better marketing decisions.

SQL Query

```
SELECT
    platform_name,
    COUNT(DISTINCT Click.click_id) AS total_clicks,
    COUNT(DISTINCT Impression.impression_id) AS total_impressions,
    (COUNT(DISTINCT Click.click_id) * 1.0 / COUNT(DISTINCT Impression.impression_id)) AS CTR
FROM SocialMediaPlatform, Campaign, Advertisement, Impression, Click
WHERE SocialMediaPlatform.platform_id = Campaign.platform_id
AND Campaign.campaign_id = Advertisement.campaign_id
AND Advertisement.ad_id = Impression.ad_id
AND Advertisement.ad_id = Click.ad_id
GROUP BY platform_name;
```

SQL Result

Result Grid		Filter Rows:	Export:	
	platform_name	total_clicks	total_impressions	CTR
▶	Facebook	8	10	0.80000
	Instagram	11	12	0.91667
	X	6	8	0.75000

Justification and Insights

The query joins the SocialMediaPlatform, Campaign, Advertisement, Impression and Click tables to link both impressions and clicks back to each platform. It calculates the click-through rate (CTR) by dividing total clicks by total impressions. This is important because CTR provides a more accurate measure of platform performance by showing how effectively each platform converts views into actual engagement.

The results show that Instagram has the highest CTR (0.9167), followed by Facebook (0.8000) and X (0.7500). This indicates that Instagram is the most effective platform in

converting advertisement views into clicks. Although Facebook has a lower CTR compared to Instagram, it still shows strong performance. X has the lowest CTR, meaning it is less effective in generating engagement. Based on this, David Jones should focus more on Instagram for future campaigns and use it as the primary platform. At the same time, the company should improve advertisement strategies on X to increase its performance.

1.8 Business concern 2

Name and Description:

Campaign Engagement Performance

This concern evaluates how effective each marketing campaign is in generating customer engagement. It measures the number of clicks received by each campaign to identify which campaigns attract the most customer interaction and interest.

SQL Query

```
#Business concern 2: Campaign Engagement Performance
SELECT campaign_name, COUNT(click_id) AS total_clicks
FROM Campaign, Advertisement, Click
WHERE Campaign.campaign_id = Advertisement.campaign_id
AND Advertisement.ad_id = Click.ad_id
GROUP BY campaign_name;
```

SQL Result

Result Grid		Filter Rows:
	campaign_name	total_clicks
▶	EOFY Sale	4
	Summer Promo	4
	Winter Sale	4
	Clearance	3
	Flash Sale	3
	Weekend Deal	2
	New Arrivals	2
	Luxury Promo	1
	Budget Deals	1
	Holiday Promo	1

Justification and Insights

The query joins the Campaign, Advertisement and Click tables to link each campaign with the clicks generated from its advertisements. It uses the COUNT(click_id) function to measure the level of engagement for each campaign. This is important because

clicks indicate customer interest, allowing the CMO to compare which campaigns are performing better and attracting more user interaction.

The results show that EOFY Sale, Summer Promo and Winter Sale all have the highest number of clicks (4 each), which means these campaigns are performing equally well. Clearance and Flash Sale come next with 3 clicks, showing moderate performance. The other campaigns have lower clicks, meaning they are less effective in attracting customer attention. Based on this, David Jones should continue focusing on the top campaigns and improve the lower-performing ones to increase engagement.

1.9 Business concern 3

Name and Description:

Customer Spending by Gender

This concern analyses customer spending patterns across different gender groups. It helps the CMO understand which customer segment contributes more revenue and supports better targeting for future campaigns.

SQL Query

```
#Business concern 3: Customer Spending by Gender
SELECT customer_gender, SUM(total_amount) AS total_spending
FROM Customer, `Order`
WHERE Customer.customer_id = `Order`.customer_id
GROUP BY customer_gender;
```

SQL Result

Result Grid			Filter Rows:
	customer_gender	total_spending	
▶	Male	1750.00	
	Female	1430.00	

Justification and Insights

The query joins the Customer and Order tables to link each customer with their purchase data. It uses the SUM(total_amount) function to calculate the total spending for each gender group. This is essential because it helps identify which customer segment contributes more to overall revenue, allowing the CMO to make better targeting decisions.

The result shows that male customers have higher total spending (1750) in comparison to female customers (1430). This indicates that male customers are contributing more

to overall revenue. To retain this high-value customer group, David Jones should design more campaigns similar to those that have previously attracted male customers. At the same time, the company should also improve its marketing strategies to increase spending from female customers and create a more balanced revenue contribution

1.10 Business concern 4

Name and Description:

Top Selling Products

This concern analyses which products are purchased the most based on quantity sold. It helps the CMO to understand customer preferences and identify high-demand products for better inventory and marketing decisions.

SQL Query

```
#Business concern 4: Top Selling Products
SELECT product_name, SUM(quantity) AS total_quantity_sold
FROM Product, OrderItem
WHERE Product.product_id = OrderItem.product_id
GROUP BY product_name;
```

SQL Result

Result Grid			Filter Rows:
	product_name	total_quantity_sold	
▶	Shirt	4	
	Shoes	4	
	Bag	3	
	Watch	2	
	Jacket	3	
	Hat	1	
	Sneakers	1	
	Dress	1	
	Jeans	1	
	Backpack	1	
	Sunglasses	1	
	Coat	1	
	Boots	1	
	T-shirt	1	
	Scarf	1	

Justification and Insights

The query joins the Product and OrderItem tables to link each product with its sales data. It uses the SUM(quantity) function to calculate the total number of units sold for each product. This is essential because it helps identify which products are most

popular among customers, allowing the CMO to focus on high-demand items for marketing and inventory decisions.

The results show that Shirt and Shoes have the highest total quantity sold (4 each), indicating that these products are the most popular among customers. Products like Bag show good demand with 3 units sold. The remaining products have lower sales, suggesting they are less popular. Based on this, David Jones should focus marketing campaigns on top-selling products like Shirt and Shoes, as well as second-highest products such as Bag to maximise sales. At the same time, the company should develop new strategies for lower-selling products, such as improving promotions, pricing, or product positioning to increase demand.

1.11 Business concern 5

Name and Description:

Order Behaviour Analysis

This concern analyses the different types of orders placed by customers (e.g., Online, In-store). It helps the CMO understand customer purchasing behaviour and identify which sales channel is most preferred.

SQL Query

```
#Business concern 5: Order Behavior Analysis
SELECT order_type, COUNT(order_id) AS total_orders
FROM `Order`
GROUP BY order_type;
```

SQL Result

Result Grid			Filter Rows:
	order_type	total_orders	
▶	Online	7	
	In-Store	4	
	Mobile	4	

Justification and Insights

The query uses the Order table to analyse customer purchasing behaviour based on order type. It uses the COUNT(order_id) function to calculate the total number of orders

for each type. This is essential because it helps identify which sales channel customers prefer, allowing the CMO to focus on improving and investing in the most popular channel.

The results show that Online orders are the highest (7), followed by In-Store and Mobile orders (4 each). This indicates that online shopping is the most preferred channel, but customers are also actively using other channels. Based on this, David Jones should conduct more online campaigns and advertisements where customers are most active to maximise engagement and sales. At the same time, the company should continue supporting In-Store and Mobile channels to maintain overall customer reach and sales performance.

1.12 Conclusion

In conclusion, the analysis shows how David Jones' marketing campaigns and customer behaviour are performing across different platforms from CMO's perspective. The results highlight that Instagram is the most effective platform based on click-through rate (CTR), meaning it converts advertisement views into clicks more efficiently than other platforms. Campaigns such as EOFY Sale, Summer Promo and Winter Sale campaigns are more effective in generating customer engagement. Likewise, it also shows that products like shirts and shoes along with male customer, contribute more to overall revenue. It is also clear that online shopping is the most preferred channel among our customers.

Based on this, we should focus more on high-performing platforms and campaigns, promote top-selling products, and continue targeting high-value customers to maximise revenue. We should also run more online campaigns and advertisements where customers are most active, especially on platforms with higher CTR. At the same time, lower-performing platforms and products should be improved through better marketing strategies.

From a scalability perspective, the database and analysis allow us to handle larger amounts of customer and marketing data as the business grows. This will help us expand our marketing strategies, improve decision-making, and support long-term growth.